A new inherent approach to solving the Collatz 3n+1 problem and its analogues

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Abstract. The article proposes a previously unseen inherent approach to the 3n+1 problem and its an±b analogues. The approach is based on the uniqueness of the decomposition of natural numbers as a product of an odd number and some power of 2. Constructive and deductive versions of proving the Collatz conjecture are given. A necessary and sufficient condition for the validity of the conjecture is determined, and possible causes of its violation are revealed. A diagrammatic method for detecting cycles is developed. A pre-classification of the an±b family algorithms with respect to the conjecture is carried out. The simplicity and productivity of the proposed approach contrasts with studies of the 3n+1 problem using sophisticated methods known in the literature. Apparently, it is worth reconsidering the current status of the Collatz conjecture. It turned out that the essence of the problem is not in numbers, but in the network structure.

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1. Introduction

The Collatz conjecture (the 3n+1 problem) is considered one of the unsolved problems in mathematics [1]. Although it is formulated simply: take any natural number, if it is even, divide by 2 while it is divisible, and if it is odd, multiply by 3 and add 1, repeat the same actions on the resulting number. The conjecture assumes that no matter what the initial number is, the result will be 1. There is a large body of research around this topic. However, various sophisticated methods have not led to decisive success [2], although since then some authors have made progress in the right direction [3], [4] and especially [5]. This article is devoted to a different — simple and natural — approach to the problem.

2. Central idea

Preliminary remark. Let's imagine a step of the Collatz algorithm somewhere inside the process: we have an odd number, multiply, add, divide, get a new odd

number to multiply again, etc. It is unclear where such iterations lead. It is absolutely necessary to include in the consideration the “finish” point (it is known that this is 1) and change the direction — to move from 1 along the ascending Collatz algorithm-in reverse (take 1, serially multiply by 2, subtract 1 and divide by 3 if divisible, get an odd number and repeat the same actions), which is more correctly considered “direct”. For brevity, we denote the original descending Collatz algorithm as “3n+1/2”, and the rules of the direct algorithm as “2n−1/3”.

As is known, any natural number can be decomposed into one odd number and 2 to some power, and in a unique way. Let's present this in a tabular form (see Fig. 1): all odd numbers are lined up in the first column, and all even numbers take their places in the rows according to the decomposition. The table as a whole can be considered generated by a two-part generator {+2, ×2} starting from 1, by analogy with the generator of the natural series — the operation {+1}. The prospects of this table for our problem are associated with two circumstances: firstly, it consists of all natural numbers without repetitions and in this sense is complete; secondly, the Collatz algorithm naturally fits into it, since it contains the rule “divide by 2 while it is divisible”.

1	2	4	8	16	32	64	128	256	512
3	6	12	24	48	96	192	384	768	1536
5	10	20	40	80	160	320	640	1280	2560
7	14	28	56	112	224	448	896	1792	3584
9	18	36	72	144	288	576	1152	2304	4608
11	22	44	88	176	352	704	1408	2816	5632
13	26	52	104	208	416	832	1664	3328	6656
15	30	60	120	240	480	960	1920	3840	7680
17	34	68	136	272	544	1088	2176	4352	8704
19	38	76	152	304	608	1216	2432	4864	9728
21	42	84	168	336	672	1344	2688	5376	10752
23	46	92	184	368	736	1472	2944	5888	11776
25	50	100	200	400	800	1600	3200	6400	12800
27	54	108	216	432	864	1728	3456	6912	13824
29	58	116	232	464	928	1856	3712	7424	14848
31	62	124	248	496	992	1984	3968	7936	15872
33	66	132	264	528	1056	2112	4224	8448	16896
35	70	140	280	560	1120	2240	4480	8960	17920
37	74	148	296	592	1184	2368	4736	9472	18944
39	78	156	312	624	1248	2496	4992	9984	19968
41	82	164	328	656	1312	2624	5248	10496	20992
43	86	172	344	688	1376	2752	5504	11008	22016
45	90	180	360	720	1440	2880	5760	11520	23040
47	94	188	376	752	1504	3008	6016	12032	24064
49	98	196	392	784	1568	3136	6272	12544	25088
51	102	204	408	816	1632	3264	6528	13056	26112
53	106	212	424	848	1696	3392	6784	13568	27136
55	110	220	440	880	1760	3520	7040	14080	28160
57	114	228	456	912	1824	3648	7296	14592	29184
59	118	236	472	944	1888	3776	7552	15104	30208

Fig. 1

Let's see how the algorithm is superposed with a table. To do this, we mark all even numbers of the form 3n+1 in it. As a result, we have the following construct: all odd numbers in the first column (highlighted in blue) serve as unique identifiers — “keys” of the rows; all even numbers of the form 3n+1 (highlighted in yellow) regularly scattered throughout the table serve as unique addresses — “pointers” for the transition to the corresponding keys. Execution of the direct algorithm 2n−1/3 on the table looks like this: multiplication by 2 is a move forward along the row, obtaining an odd number from an even number of the form 3n+1 is a transition from the pointer to the key that begins another row. For the original algorithm, it's the same in reverse. (Further, we will use the short notations: Table, key, pointer.) In essence, the resulting Table is an adequate visualization of the network of natural numbers and the operations on them implicitly defined by Collatz.

3. Constructive proof

As already noted, the Collatz conjecture is more correctly considered an inverse problem. Let us formulate an equivalent direct problem: it is required to prove that any finite natural number is reachable from the starting point (number 1) by recurrently applying the rules of the direct algorithm 2n−1/3. On the Table, this means that any cell is reachable from cell 1 by the 2n−1/3 algorithm, or, since the reachability of any even number is a trivial consequence, any row of the Table starting with an odd number is reachable.

Due to the completeness of the Table and the one-to-one correspondence of the key – pointer pairs (mapping n – 3n+1), there is the only transition to any key from a pointer in another row. This entails the connectivity of all rows, which can only be prevented by a significant "flaw" in the network of keys and pointers. But checking the 3n+1 pointers reveals only one trivial cycle of row 1 onto itself (1-2-4-1). For our proof, such a cycle is both "harmless" since it does not capture other rows, and "useful" since it stops the descending algorithm 3n+1/2 at the number 1, giving the ascending algorithm a starting point.

In the absence of violations of the network of keys and pointers, the statement about the single-connectivity of all rows and the only recursive progenitor — the number 1 in row 1 — is true. This kind of connectivity ensures the unstoppability of the general forward movement along the Table — visits to rows and multiplying transitions from pointer cells (yellow) to key cells (blue) of new and new rows. In this case, the strategy of movement along the Table must be "diagonal" to avoid going to infinity along the rows. Fig. 2 illustrates the beginning of the process: the 2n−1/3 algorithm started from 1 and stopped at the transition from pointer 58 to key 19.

1	2	4	8	16	32	64	128	256	512
3	6	12	24	48	96	192	384	768	1536
5	10	20	40	80	160	320	640	1280	2560
7	14	28	56	112	224	448	896	1792	3584
9	18	36	72	144	288	576	1152	2304	4608
11	22	44	88	176	352	704	1408	2816	5632
13	26	52	104	208	416	832	1664	3328	6656
15	30	60	120	240	480	960	1920	3840	7680
17	34	68	136	272	544	1088	2176	4352	8704
19	38	76	152	304	608	1216	2432	4864	9728
21	42	84	168	336	672	1344	2688	5376	10752
23	46	92	184	368	736	1472	2944	5888	11776
25	50	100	200	400	800	1600	3200	6400	12800
27	54	108	216	432	864	1728	3456	6912	13824
29	58	116	232	464	928	1856	3712	7424	14848
31	62	124	248	496	992	1984	3968	7936	15872
33	66	132	264	528	1056	2112	4224	8448	16896
35	70	140	280	560	1120	2240	4480	8960	17920
37	74	148	296	592	1184	2368	4736	9472	18944
39	78	156	312	624	1248	2496	4992	9984	19968
41	82	164	328	656	1312	2624	5248	10496	20992
43	86	172	344	688	1376	2752	5504	11008	22016
45	90	180	360	720	1440	2880	5760	11520	23040
47	94	188	376	752	1504	3008	6016	12032	24064
49	98	196	392	784	1568	3136	6272	12544	25088
51	102	204	408	816	1632	3264	6528	13056	26112
53	106	212	424	848	1696	3392	6784	13568	27136
55	110	220	440	880	1760	3520	7040	14080	28160
57	114	228	456	912	1824	3648	7296	14592	29184
59	118	236	472	944	1888	3776	7552	15104	30208

Fig. 2

From the single-connectivity of all rows, the only recursive progenitor and the countability of the diagonal strategy, it follows that all finite natural numbers are reachable from the starting point (number 1) using the 2n−1/3 algorithm in a finite number of steps — the direct conjecture is proven. The validity of the Collatz conjecture itself (the inverse one) is a consequence: any natural number n will be reduced by the 3n+1/2 algorithm to 1 in a finite number of steps.

For understanding, let us give the following interpretation of the direct algorithm: reachability of any number means successful branching traversal of all natural numbers, i.e. traversal of all cells of our infinite Table. The latter is already easy to translate into the language of topology (graph theory): the rows of the Table are branches with even nodes, transitions pointer to key are branching, the starting point is the root of the tree, then reachability of any number means successful traversal of the infinite binary rooted tree with numbering of forks all odd numbers according to the scheme specified by the algorithm. Accordingly, the interpretation of the result of the work of the Collatz algorithm: successful descent from any branch of the tree to its only root. The root is understood as the end point of the descending algorithm, which is also the starting point of the ascending algorithm, if such exists.

4. Threats to connectivity

To figure it out with possible violations of the connectivity of the network of keys and pointers, we turn to analogues of the Collatz algorithm. The 3n+1 rule can be changed; any formula $m = f(n)$ or mapping $m = F(n)$ is suitable for this under one condition: the set of all odd numbers n is mapped one-to-one onto the set of even numbers m. Another formula will specify a different scheme for placing pointers in the Table. The simplest suitable option is the family of formulas $m = an + b$, where the odd parameters $a = 1, 3, 5\ldots$ and $\pm b = 1, 3\ldots (2a - 1)$. Let us again introduce short notations for the algorithms: "an+b/2" (or simply "an+b") for reverse descending, "2n−b/a" for direct ascending.

Let's start with the simplest algorithm 1n+1/2 (see Fig. 3): pointers "fill" the entire Table without gaps, moving diagonally from the only root 1. The maximum filling density indicates high efficiency (speed of traversing natural numbers) of the direct algorithm 2n−1/1: each pointer is a branching, and they come in a row, not with gaps. There is no doubt that this algorithm does not encounter cycles and traverses all natural numbers without any problems, although it also generates a "interstriping" of active rows gradually going down the Table, and it also has its own trivial cycle (1-2-1).

1	2	4	8	16	32	64	128	256	512
3	6	12	24	48	96	192	384	768	1536
5	10	20	40	80	160	320	640	1280	2560
7	14	28	56	112	224	448	896	1792	3584
9	18	36	72	144	288	576	1152	2304	4608
11	22	44	88	176	352	704	1408	2816	5632
13	26	52	104	208	416	832	1664	3328	6656
15	30	60	120	240	480	960	1920	3840	7680
17	34	68	136	272	544	1088	2176	4352	8704
19	38	76	152	304	608	1216	2432	4864	9728
21	42	84	168	336	672	1344	2688	5376	10752
23	46	92	184	368	736	1472	2944	5888	11776
25	50	100	200	400	800	1600	3200	6400	12800
27	54	108	216	432	864	1728	3456	6912	13824
29	58	116	232	464	928	1856	3712	7424	14848
31	62	124	248	496	992	1984	3968	7936	15872
33	66	132	264	528	1056	2112	4224	8448	16896
35	70	140	280	560	1120	2240	4480	8960	17920
37	74	148	296	592	1184	2368	4736	9472	18944
39	78	156	312	624	1248	2496	4992	9984	19968
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45	90	180	360	720	1440	2880	5760	11520	23040
47	94	188	376	752	1504	3008	6016	12032	24064
49	98	196	392	784	1568	3136	6272	12544	25088
51	102	204	408	816	1632	3264	6528	13056	26112
53	106	212	424	848	1696	3392	6784	13568	27136
55	110	220	440	880	1760	3520	7040	14080	28160
57	114	228	456	912	1824	3648	7296	14592	29184
59	118	236	472	944	1888	3776	7552	15104	30208

Fig. 3

A trivial cycle (or 1-cycle, since it is limited to one row of the Table and contains only one key, perhaps it's more correct to call it a “loop”) in the an±b family is a common phenomenon. It owes its origin to the presence in the Table of a pointer to the root above the root itself in the same row. The equation for finding trivial cycles is $2^p \times n = an + b$ ($p = 1, 2, 3\ldots$), it may have more than one solution. If a 1-cycle exists, it designates the root row, because it is impossible to get to this row by the direct algorithm except by starting from it. A trivial cycle does not pose a threat to single-connectivity until there is a second root.

Another way to designate the root row is due to the absence of a pointer to the root in the Table, which is the case for all an+b algorithms if $b < 0$ and $|b| \geq a$. Similar to the previous one, it is impossible to get to such a row by the direct algorithm except by starting from it, and there is no threat to single-connectivity until there is a second root.

Topologically, the presence of two roots means that the single tree has split into two unconnected ones, each with its own root, and the descent “according to Collatz” will lead to either one or the other root. For algorithms with two roots, the proof proposed above (without adequate modification) does not work.

The opposite situation is possible, when the network of keys and pointers has no designated root, as, for example, in the case of 9n+1. With perfect single-connectivity, from any point of such a network there are moves by both the ascending and descending algorithms. This means that the algorithm has no start or finish point, the recurrence is infinite in both directions, and the Collatz conjecture is obviously incorrect.

The 5n+1/2 algorithm (see Fig. 4) revealed the following complexity — a non-trivial cycle of two rows (1-2-4-8-16-3-6-1). In this case, the proof proposed above can still work, since the 2-cycle contains the root 1. But in the general case, a non-trivial cycle breaks the single-connectivity of the network of keys and pointers.

1	2	4	8	16	32	64	128	256	512
3	6	12	24	48	96	192	384	768	1536
5	10	20	40	80	160	320	640	1280	2560
7	14	28	56	112	224	448	896	1792	3584
9	18	36	72	144	288	576	1152	2304	4608
11	22	44	88	176	352	704	1408	2816	5632
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17	34	68	136	272	544	1088	2176	4352	8704
19	38	76	152	304	608	1216	2432	4864	9728
21	42	84	168	336	672	1344	2688	5376	10752
23	46	92	184	368	736	1472	2944	5888	11776
25	50	100	200	400	800	1600	3200	6400	12800
27	54	108	216	432	864	1728	3456	6912	13824
29	58	116	232	464	928	1856	3712	7424	14848
31	62	124	248	496	992	1984	3968	7936	15872
33	66	132	264	528	1056	2112	4224	8448	16896
35	70	140	280	560	1120	2240	4480	8960	17920
37	74	148	296	592	1184	2368	4736	9472	18944
39	78	156	312	624	1248	2496	4992	9984	19968
41	82	164	328	656	1312	2624	5248	10496	20992
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45	90	180	360	720	1440	2880	5760	11520	23040
47	94	188	376	752	1504	3008	6016	12032	24064
49	98	196	392	784	1568	3136	6272	12544	25088
51	102	204	408	816	1632	3264	6528	13056	26112
53	106	212	424	848	1696	3392	6784	13568	27136
55	110	220	440	880	1760	3520	7040	14080	28160
57	114	228	456	912	1824	3648	7296	14592	29184
59	118	236	472	944	1888	3776	7552	15104	30208

Fig. 4

System of equations for finding 2-cycles: $2^p \times n_1 = an_2 + b$, $2^q \times n_2 = an_1 + b$ ($p = 1, 2, 3\ldots$, $q = 1, 2, 3\ldots$). Similarly, to search for 3-cycles: $2^p \times n_1 = an_2 + b$, $2^q \times n_2 = an_3 + b$, $2^r \times n_3 = an_1 + b$ ($p = 1, 2, 3\ldots$, $q = 1, 2, 3\ldots$, $r = 1, 2, 3\ldots$), etc.

Solutions of such systems of equations have to be selected. And such computer calculations have already been partially (for an+1) done [6]. It turned out that non-trivial cycles are extremely rare. The program found only five: with $a = 5$, three cycles — $(1, 3)$, $(13, 83, 33)$ and $(17, 27, 43)$; with $a = 181$, two cycles — $(27, 611)$ and $(35, 99)$.

There is an explanation for this. The main determinant of the matrix of the system of linear equations for finding cycles always has the form $D = 2^z - a^d$, where d is the cycle size (matrix size), z is the sum of powers of 2 over all d steps of the cycle. For an integer solution to exist, D must be a common divisor for all d additional determinants. This is a very strong requirement, making cycles unrealistic and even more unrealistic the larger the parameter a and the cycle size d. Therefore, 2-cycles with $a = 181$ $(D = 2^{15} - 181^2 = 7)$ are a rare "lucky" accident.

5. Diagrammatic method

As follows from the previous section, the most favorable value for the existence of cycles is $D = \pm 1$ (plus for $b > 0$, minus for $b < 0$). Then, as a heuristic criterion for the closing of a d-cycle, we can take $2^z \approx a^d \pm 1$. Let's see how this criterion is fulfilled for small cycles in the younger representatives of the an±b family, using the following table (the values closest to the criterion are highlighted in color):

a	3	5	7	9	...	181
a±1	4/2	6/4	8/6	10/8	...	182
a^2±1	10/8	26/24	50/48	82/80	...	32762
a^3±1	28/26	126/124	344/342	730/728	...	5929742
2^z	$2=2^1$ $4=2^2$ $8=2^3$ 16 $32=2^5$ 64 $128=2^7$ 256 512 1024 ...					$32768=2^{15}$

The beginning of the line a±1 correlates well with the presence of the trivial 1-cycles already considered above in the corresponding algorithms: 3n±b, 5n−b, 7n+b and 9n−b. In the next two lines, the criterion predicts four more "cyclo-dangerous" algorithms: 2-cycles in 3n−b and 5n+b, 3-cycles in 3n+b and 5n+b. A 2-cycle in the "exotic" 181n+1 mentioned in the previous section is quite probable. But the criterion only allows us to assume the existence of cycles, and we would like to have a tool for calculating them, giving a complete final answer.

Let's look again at our Table for the case of the 5n+1/2 algorithm (see Fig. 4) and turn the 3-cycles discovered here (see section 4) into working diagrams. The 3-cycle closing criterion $2^7 \approx 5^3 + 1$ and the fact that the pointers in the Table in the first two columns always lead to keys backwards, and in the third and further — to keys forwards, allow us to identify all possible 3-cycle configurations for $a = 5$ — there are five of them: 5-1-1, 4-2-1, 4-1-2, 3-3-1, and 3-2-2 (see Fig. 5). The digital notation of the diagrams reflects the distribution of the power $2^7 = 2^p \times 2^q \times 2^r$ into three steps of the cycle. In the diagram, the turning points on the left are keys, the turning points on the right are pointers, the lines with arrows are transitions from pointer to key, and the simple lines are series of doublings.

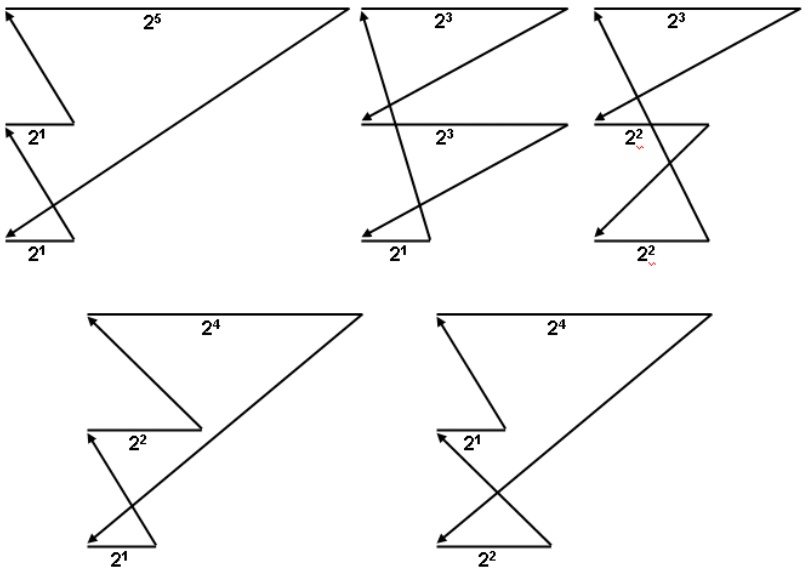

Fig. 5

The jump to diagrams greatly simplifies the calculation of cycles, since three equations with six unknowns turn into a system of three linear equations with three unknowns (for each of the five diagrams). For example, for the 5-1-1 diagram: $2^5 \times n_1 = 5n_2 + 1$, $2^1 \times n_2 = 5n_3 + 1$, $2^1 \times n_3 = 5n_1 + 1$. Solving in the standard way, we obtain the already known cycle $(13, 83, 33)$, and the 3-3-1 diagram gives the second cycle $(17, 27, 43)$. The remaining diagrams 4-2-1, 4-1-2 and 3-2-2 "have" non-integer solutions, which reflects the fact that there are no 3-cycles corresponding to them. Similarly, using the diagrammatic method, any other algorithm of the form an±b is checked for the presence of cycles.

6. Catalog and classification

Since the main determinant of the matrix of the system of linear equations for finding cycles $D = 2^z - a^d$ is always nonzero, the solution is unique, which means that there can be no more than one(!) cycle of each configuration. The number of allowed diagrams (configurations) corresponding to potential cycles, given parameters a, d and a meaningful value of z, is finite(!). This allows a complete cataloging of cycles using the diagrammatic method. The following table contains all the cycles found (up to d = 4) in the younger representatives (up to a = 9) of the an±b family (sporadic cycles for some values of b are highlighted in color, the keys included in the cycle are listed in parentheses, and the corresponding diagrams are additionally provided for 2- and 3-cycles).

Catalog of cycles	a=1 b=1	a=3 b=1,3,5	a=5 b=1,3…7,9	a=7 b=1…13	a=9 b=1…17	a=181 b=1
1-cycles	1n+1 (1)	3n+b (b) 3n−b (b) 3n+5 (1)	5n−b (b) 5n+3 (1) 5n+9 (3) 5n−3 (1) 5n−9 (3)	7n+b (b) 7n+9 (1) 7n−3 (1) 7n−5 (1) 7n−9 (3)	9n−b (b) 9n+7 (1) 9n−5 (1) 9n−7 (1) 9n−15 (3)	–

2-cycles	–	3n−b 2-1 (5b,7b)	5n+b 4-1 (b,3b) 5n+7 3-2 (9,13) 5n−9 3-1 (7,13)	–	–	181n+1 12-3 (27,611) 9-6 (35,99)
3-cycles	–	3n+5 3-1-1 (19,49,31) 2-2-1 (23,29,37)	5n+b 5-1-1 (13b,83b,33b) 3-3-1 (17b,27b,43b) 5n+3 4-2-1 (43,137,109) 4-1-2 (53,169,67) 3-2-2 (61,97,77) 5n+9 4-2-1 (129,411,327) 4-1-2 (159,507,201) 3-2-2 (183,291,231)	–	–	–
4-cycles	–	–	–	–	–	–

Given the above (starting with section 4), we now have everything we need to pre-classify the representatives of the an±b family with respect to the Collatz conjecture, i.e. to indicate which of them can converge to some number (like 3n+1 converging to 1), and which ones will not, and why.

Algorithms for which the Collatz conjecture is true correspond to networks of keys and pointers, which are simply connected rooted trees, the only root (starting point) of which is designated: either by the absence of a pointer to it in the Table, or by the presence of such a pointer in the root cycle. This is the necessary and sufficient condition for the validity of the Collatz conjecture for the 3n+1 algorithm and its an±b analogues. The list of causes for violating this condition is: (1) absence of a root, (2) non-uniqueness of the root, (3) non-root cycle, or their combinations (for more details about the cause (4), see Addendum 21.10.2024). In the following table, all younger representatives (up to a = 9) of the an±b family are divided into two classes based on the convergence criteria of the algorithm (the reason for the invalidity of the conjecture is given in parentheses).

Collatz conjecture	a=1 b=1	a=3 b=1,3,5	a=5 b=1,3…7,9	a=7 b=1…13	a=9 b=1…17	a=181 b=1
true?*	1n+1 1n−1	3n+1 3n+3	5n−1	7n+b 7n−b	9n+7 9n−1 9n−3	

false		3n+5 (2,3) 3n−1 (3) 3n−3 (2,3) 3n−5 (2,3)	5n+1 (3) 5n+3 (2,3) 5n+5 (3) 5n+7 (3) 5n+9 (2,3) 5n−b (2) 5n−9 (2,3)	7n+9 (2) 7n−1 (1) 7n−9 (2)	9n+b (1) 9n−b (2)	181n+1 (1,3)

*Preliminary, according to the presence of roots and cycles (see Addendum 21.10.2024)

7. Deductive proof

For greater certainty, let us rewrite in a deductive style the constructive proof of the Collatz conjecture presented above (see section 3). The formulation of the problem is the same: it is required to prove that any finite natural number is reachable from the starting point (number 1) by recurrently applying the rules of the direct algorithm 2n−1/3.

The proof is based on one equation and its two properties. Let us write the main equation of the 2n−1/3 algorithm (hereinafter EQ) for the transition from the previous odd number n_{i-1} to the next odd number n_i:

$2^p \times n_{i-1} = m_i = 3n_i + 1$

where left and right are even expressions, m_i is an even number (the one that is divided by 2 in the original formulation of the conjecture), p = 1, 2, 3… is the exponent of 2. Note that for a known n_{i-1} there is an infinite (due to enumeration over p) series of solutions — many pairs $|m_i, n_i>$, for a known n_i there is always a single solution — one pair $<n_{i-1}, m_i|$.

The meaning of the EQ is as follows: in the left expression one recognizes the only possible decomposition of any even number as a product of an odd and some power of 2 (hereinafter D2), on the right — one of the formulas for the one-to-one converting of any odd number into an even number (hereinafter OE). The domains of definition on the left and right coincide — these are all odd numbers. Taken together, the EQ is a two-part recurrent formula that connects two odd ones through an even "intermediary". We will show that the application of this formula allows us to recursively obtain from 1 any finite odd number (direct problem).

Let's check the EQ for trivial cycles: $n_{i-1} = n_i = n$, $2^p \times n = 3n + 1$, $(2^p - 3) \times n = 1$, the only solution n = 1 confirms the correct choice of the starting point (number 1), then n_{i-1} and n_i are always different. This algorithm does not have any non-trivial cycles that complicate the proof (see section 6).

Let's start applying the EQ, starting with $n_0 = 1$: at the first step $2^p \times n_0 = m_1 = 3n_1 + 1$ we obtain the first series of pairs $|m_1, n_1>$, i.e. the process branches. We ignore the first solution in the series $n_0 = n_1 = 1$ (as static), take the other n_1 for substitution in the second step $2^p \times n_1 = m_2 = 3n_2 + 1$, we get many series of pairs $|m_2, n_2>$, etc.

Now the main thing in the proof! At each step of applying the EQ, due to the first property — the uniqueness of D2, only unique new even numbers m_i will be obtained, belonging to their type $2^p \times n_{i-1}$, and then, as a consequence of the second property — the one-to-one of OE, only unique new odd numbers n_i will be obtained from m_i. This means that the recurrence builds branching chains without any repetition (or cycle) — such a possibility is excluded.

In this case, the array of all recurrently received $[n_i]$ will be steadily replenished, but not in order, but with interstriping, the boundary of which, determined by the first odd not yet received in ascending order, will steadily shift to larger numbers.

In order not to go into infinity of series, we use a "diagonal" strategy of enumerating multiplying solutions for recurrent substitution, alternating in a circle progress through all series with a preference for smaller n_i. As is known, the diagonal strategy is countable. This means that any finite odd number will end up in the array $[n_i]$ after some finite number of steps.

The direct conjecture for the $2n-1/3$ algorithm is proven.

The proof of the inverse conjecture for the $3n+1/2$ algorithm is a consequence. In this case, the EQ simply turns around:

$2^p \times n_{i+1} = m_i = 3n_i + 1$

We take any initial odd number, apply the EQ, each step of the inverse recurrence gives a single pair $<n_{i+1}, m_i|$, i.e. the process does not branch. The resulting n_i are arranged in a linear chain, always leading to the number 1, as the starting point of the direct recurrence.

The Collatz conjecture is proven.

If we compare our two proofs (deductive and constructive), we observe complete equivalence: the decomposition D2 is visualized in the Table, even m_i are pointers, odd n_i are keys of the Table, the formula OE is the key – pointer relationship… But the constructive approach is more productive, since it provides a better understanding of the problem.

8. Conclusion

Summary of the results. A previously unseen inherent approach to the $3n+1$ problem and its $an\pm b$ analogs is proposed. It is based on the uniqueness of the decomposition of natural numbers as a product of an odd number and some power of 2. The constructiveness of the approach lies in combining the algorithm with a decomposition Table and operates on a network of keys and pointers. Constructive and deductive versions of proving the Collatz conjecture are given. A necessary and sufficient condition for the validity of the conjecture for the original algorithm and analogues is the single-connectivity of the network of keys and pointers in the presence of a single root. Possible causes for violating this condition are revealed: absence of a root, non-uniqueness of the root, non-root cycle, other network fragmentation (see Addendum 21.10.2024). A diagrammatic method for detecting cycles is developed, which reduces the problem to a trivial one. The known results of numerical calculations coincide with those obtained by this method. A catalog of cycles is compiled and a pre-classification of younger representatives (up to $a = 9$) of the $an\pm b$ family with respect to the conjecture is carried out.

The simplicity and productivity of the proposed approach contrasts with studies of the $3n+1$ problem using sophisticated methods known in the literature. Apparently, it is worth reconsidering the current status of the Collatz conjecture. It turned out that the essence of the $3n+1$ problem is not in the properties and behavior of numbers as such, but in the structure of the network of numbers and operations on them. The seemingly inexplicable "wanderings" are not actually random, but are driven by the network. It seems that now it will be possible to answer many difficult questions [2] posed over the years of studying the problem.

Finally, let us note the main ideas that can help in this. (1) The construction combining the algorithm with the decomposition of a natural series into a Table can work as a standard tool for studying at least all two-part algorithms containing multiplication and (or) division by 2 and the transition even – odd. The success of this decomposition lies in its uniqueness and completeness — there is no other

like it, so some algorithms are “luckier” than others. (2) However, it may potentially be a fruitful idea to consider a broader class of two-part algorithms as a heterogeneous network of numbers and operations on them, where numbers are of different types, according to their role in the operations. Auxiliary idea about the connection with topology: two-part algorithms should correspond to two-dimensional objects — infinite graphs.

Addendum 30.09.2024

Let's return to the question of hypothetical large cycles in algorithms of the $an \pm b$ family: do they exist? The system of linear equations for finding cycles and the corresponding matrix are simple, due to which the determinant D and additional determinants D_i have a special form. Let's show it using the example of a 5-cycle for the algorithm $an+b$:

$2^p \times n_1 - an_2 = b$

$2^q \times n_2 - an_3 = b$

$2^r \times n_3 - an_4 = b$

$2^s \times n_4 - an_5 = b$

$2^t \times n_5 - an_1 = b$

where p, q, r, s, t = 1, 2, 3… are powers of 2 over five steps of the cycle.

$D = 2^p 2^q 2^r 2^s 2^t - a^5$

$D_1 = (2^q 2^r 2^s 2^t + 2^r 2^s 2^t a + 2^s 2^t a^2 + 2^t a^3 + a^4)b$

$D_2 = (2^r 2^s 2^t 2^p + 2^s 2^t 2^p a + 2^t 2^p a^2 + 2^p a^3 + a^4)b$

$D_3 = (2^s 2^t 2^p 2^q + 2^t 2^p 2^q a + 2^p 2^q a^2 + 2^q a^3 + a^4)b$

$D_4 = (2^t 2^p 2^q 2^r + 2^p 2^q 2^r a + 2^q 2^r a^2 + 2^r a^3 + a^4)b$

$D_5 = (2^p 2^q 2^r 2^s + 2^q 2^r 2^s a + 2^r 2^s a^2 + 2^s a^3 + a^4)b$

Since the parameters a and b are odd, D and all D_i are odd too. For a 5-cycle to exist, the system must have an integer solution, which means that D must be a common divisor of all D_i. This is possible in the following cases:

1) $D = \pm 1$ and the signs of D and D_i coincide, our heuristic criterion $2^z \approx a^d \pm 1$ (see Section 5) turned out to be just this case,

2) D is a divisor of the parameter b, in the special case $D = b$,

3) D is a common divisor for all D_i, but not case 2), i.e. the divisor affects the expressions in parentheses.

If we compare with the catalog of cycles (see Section 6):

case 1) is responsible for trivial 1-cycles in the algorithms $1n+1$, $3n \pm b$, $5n-b$, $7n+b$, $9n-b$ and non-trivial 2-cycles in $3n-b$,

case 2) corresponds to sporadic cycles for some values of the parameter b (highlighted in color in the catalog),

case 3) is rare non-trivial 2-cycles in $5n+b$, $181n+1$ and 3-cycles in $5n+b$, there are no more such “lucky” cycles here.

For cases 1) and 2), the cycle configuration (diagram) does not matter, only the sum of the powers $z = p + q + r + s + t$ is important. The feasibility of case 3), on the contrary, is determined by the existence of an integer solution for a specific cycle configuration, and the distribution of powers of 2 over the five steps of the cycle p-q-r-s-t is the previously used designation of the corresponding cycle diagram.

The feasibility of case 1) depends entirely on the existence of solutions to the equation $2^z - a^d = \pm 1$ in integers. Such solutions exist only for $d = 1$ and, as an exception, for $d = 2$ and $a = 3$ (see table in Section 5). For $d \geq 3$, there are no cycles of this type for any values of the parameter a.

Note. The above statement is true due to the proof of Catalan's conjecture (or Mihăilescu's theorem), the case $d = 2$ and $a = 3$ corresponds exactly to the unique solution $3^2 - 2^3 = 1$.

Case 3) is unrealistic for another reason: as the size of the cycle grows, the number and magnitude of additional determinants D_i grows, and all of them must be divisible by the determinant D. And if there is at least one prime number among D_i, then divisibility is completely sad. That is why there are so few cycles of this type — two for $d = 2$ and one for $d = 3$. Due to [7], another unique 7-cycle was found in the 3n−1 algorithm (17, 91, 61, 41, 55, 37, 25), $D = 2^{11} - 3^7 = -139$.

Thus, we are left with only one type of large cycles — these are sporadic cycles 2). They are not difficult to "construct" for any cycle size (value of d). Let's show this using the example of 5-cycles in the 3n+13 algorithm. Why do we take $b = 13$? Because in our example $D = 2^z - a^d = 2^8 - 3^5 = 13$ (for $z = 8$). Calculating D_i, we sequentially obtain the entire series of 5-cycles, starting with the diagram 4-1-1-1-1 — cycle (211, 1121, 743, 491, 323), then 3-2-1-1-1 — cycle (227, 601, 797, 527, 347), etc. until the end of the enumeration of allowed diagrams (remember that for $a = 3$, forward moves 4, 3, 2 are allowed, backward move only 1). In this way, we can calculate cycles of any size for any value of a, but with one selected value $b = D$.

In the end, if we do not take into account the somewhat "artificial" sporadic cycles, we can conclude that the an±b algorithms do not have large cycles (starting with $d = 8$).

Addendum 21.10.2024

Returning to threats to single-connectivity. In Section 4, the last and most obscure threat was left unconsidered. This is the general "degradation" of the network of keys and pointers for an±b algorithms compared to the "ideal" network for 1n±1 algorithms as the parameter a grows. Degradation is fraught with a critical disruption of network connectivity. Its main cause is the scaling of the network of keys and pointers by one of the components of the two-part algorithm 3n+1, 5n+1, 7n+1, etc., while the underlying network (table of natural numbers) remains unchanged. The degradation is also caused by the appearance of empty rows in the Table for 3n+1, 5n+1, 7n+1, etc., which do not contain pointers.

To understand what the threat to our proof is, let us again turn to the simplest algorithm 1n+1 (see Fig. 3). Its peculiarity is that all pointers lead only to keys forward, since $2n > 1n + 1$. This allows us to prove the completeness of the traversal of the entire Table by this algorithm, i.e. the reachability of all odd and, as a consequence, all even numbers. It is simply necessary to sequentially activate those pointers that lead to the next ascending keys — such a diagonal strategy eliminates the occurrence of "interstriping" of active rows and in a finite number of steps reaches any natural number. Since all steps are deterministic, the completeness of the Table traversal and the equivalent single-connectivity of the entire 1n+1 network are proven.

Not so in the case of the 3n+1 algorithm. Here, in the first column of the Table, pointers appear that lead backward (see Fig. 2), since $2n < 3n + 1 < 4n$. Because of this, interstriping is inevitable with any version of the diagonal strategy for

traversing the Table. If the network is simply connected, interstriping is not a problem, for any key value (odd number) it is overcome in a finite number of steps. If the network is not simply connected, the persisting interstriping will reveal the fragmentation of the network, i.e. disintegration into separate endless pieces. A logical scenario for the disintegration of a network with a single root is a dividing into a connected fragment with a root and a connected rootless fragment, one or more. Unlike 1n+1, here the single-connectivity of the entire 3n+1 network is a necessary condition for the completeness of the Table traversal.

Thus, the last identified threat to the above proof of the Collatz conjecture for the 3n+1 algorithm (see Sections 3 and 7) is the hypothetical fragmentation of the network, which disrupts the single-connectivity, as well as extra roots and cycles. Traditionally, "divergences" are also called a threat, but as we found out, this can hide either simply connected rootless networks or disintegrated networks with a rootless fragment. All empirical data suggest that in the 3n+1 case there is no fragmentation, i.e. this threat was also not realized.

This addendum was born after a colleague's remark about the discrepancy of the 7n+1 case with the proposed classification (see Section 6). Numerical calculations show that in this case the threat was just realized — the 7n+1 network (formal analogue of 3n+1) is fragmented, most likely according to the scenario described above. Therefore, this classification should not be understood as final, it was compiled according to the criteria of the presence of roots and cycles without taking into account the possibility of network disintegration. It would be interesting to know how it really is: are the networks of the remaining candidates for Collatz convergence fragmented — 5n−1, 7n+1 (apparently yes), 9n−1? I hope we find out soon…

Addendum 15.12.2024

Final remarks and conclusions. For even greater clarity, we reformulate the original problem. By sequential substitution it is not difficult to obtain an infinite series of equations connecting two keys (odd numbers) as the number of steps increases according to the ascending algorithm "2n−b/a" (corresponding to the descending "an+b/2"):

$a^1n_1 = 2^pn_0 - b$

$a^2n_2 = 2^p2^qn_0 - 2^qb - ba$

$a^3n_3 = 2^p2^q2^rn_0 - 2^q2^rb - 2^rba - ba^2$

$a^4n_4 = 2^p2^q2^r2^sn_0 - 2^q2^r2^sb - 2^r2^sba - 2^sba^2 - ba^3$

etc. in an obvious way, where p, q, r, s… are powers of 2 in the algorithm steps, n_0 is the starting key. It was not possible to get rid of the recursion completely, but the intermediate steps were absorbed.

Note. Let us note that by equating $n_i = n_0$ (i = 1, 2, 3, 4…), we come to another formulation of the condition for the existence of cycles in the network. But the approach to cycles used in this article is more productive, since it allows one to use the technique of solving systems of linear equations. By the way, it also shows the falsity of the still-existing myths about the possibility of "infinite cycles".

Somewhat simpler is the equivalent series of equations relating pointers (even numbers) to the starting key:

$m_1 = 2^pn_0$

$a^1m_2 = 2^p2^qn_0 - 2^qb$

$a^2m_3 = 2^p2^q2^rn_0 - 2^q2^rb - 2^rba$

$a^3m_4 = 2^p2^q2^r2^sn_0 - 2^q2^r2^sb - 2^r2^sba - 2^sba^2$

These two series of equations are an expression of *local connectivity* and are valid for any network topology generated by an algorithm of the an±b family. But the problem with the Collatz conjecture is that it is necessary to prove the *global single-connectivity* of the network of all natural numbers with a single root, which corresponds to the tree topology. This means that for any odd number (n_{odd}), with the obligatory starting from the root ($n_0 = b$), there must be an equation in the series that has a solution — a finite combination of p, q, r, s..., defining the path from b to n_{odd}. Similarly, in the series of equations with pointers, but not for any even number, but only for $m_{even} = an_{odd} + b$.

Note: If the root is generated by a trivial cycle, there is a nuance: the solution will not be unique, since subsequent equations accumulate all the solutions of the previous ones. Therefore, the desired "minimal" equation will be the one whose first step 2^p is not trivial.

Is it possible to prove that the network is simply connected (in fact, the main and last obstacle in proving the Collatz conjecture) in this formulation?

In the simplest case 1n±1 — yes, very easy. Above (see Addendum 21.10.2024) this has already been done in a constructive way. It is also obvious that if we substitute the parameter values into the equations, we get a series of recognizable formulas for generating all odd and even numbers using powers of 2 and 1, for example:

$n_4 = 2^p2^q2^r2^s \mp 2^q2^r2^s \mp 2^r2^s \mp 2^s \mp 1$

$m_4 = 2^p2^q2^r2^s \mp 2^q2^r2^s \mp 2^r2^s \mp 2^s$

The 3n+1 algorithm case is more complicated. Substituting the parameter values gives a series of equations that are mixed expansions with powers of 2 and 3. But the most unpleasant thing is the power of 3 on the left side, which increases with each step:

$3^4n_4 = 2^p2^q2^r2^s - 2^q2^r2^s - 2^r2^s3 - 2^s3^2 - 3^3$

$3^3m_4 = 2^p2^q2^r2^s - 2^q2^r2^s - 2^r2^s3 - 2^s3^2$

With such a recurrent-dynamic nature of the network connectivity equations, it is logical to believe that a *correct proof* can only be by recurrence with completion or external to the Collatz algorithm itself. Among the vast mountains of publications around the 3n+1 problem, we were able to find one acceptable proof, obtained by independent researcher *Leszek Mazurek in 2021* [8].

It would not be amiss to add here why other approaches, in our opinion, will not lead to success. Proof by induction (due to the dynamism of the above equations and non-linearity — branching of the network) and proof based on certain properties of numbers (since behind their evolution there is nothing but a relation to the network connectivity equations) are unpromising; one should not expect a proof from within the recursion — following the algorithm (most likely it will fall into the trap of the implicit assumption that the network is simply connected), etc.

The next candidate for Collatz convergence in our table (see Section 6) is the 3n+3 algorithm. Substituting the root value $n_0 = b$ into the series of equations, we get:

$a^4n_4 = 2^p2^q2^r2^sb - 2^q2^r2^sb - 2^r2^sba - 2^sba^2 - ba^3$

$a^3m_4 = 2^p2^q2^r2^sb - 2^q2^r2^sb - 2^r2^sba - 2^sba^2$

From this we can see that all solutions for the case 3n+1 are solutions for 3n+3, but with tripled values of the keys (n_i) and pointers (m_i), forming the skeleton of this network, sufficient for successful traversal of the entire Table (see section 3). Thus, for 3n+3 the Collatz conjecture is true simultaneously with 3n+1.

The 3n+5 algorithm could have kept them company, but it was "let down" by three cycles (see Section 6), as a result of which this network must split into three separate trees with cyclic roots at the base of each. However, proving this fact will be even more difficult than the single-connectivity of 3n+1.

Finally, the algorithms 5n−1, 7n+1, 9n−1…, which remain in question (see Section 6). Most likely, starting with a = 5, the networks break up into a rooted tree and rootless fragments. The reason for the disintegration is that as the parameter a increases, the above network connectivity equations become "deficient" and no longer ensure the generation (under the condition of starting from a single root) of all odd numbers by combining powers of twos and 5, 7, 9. Moreover, the effect is cooperative: the numbers fall out not one at a time, but in infinite "collectives".

It is interesting that the single-connectivity of a rootless network is unprovable in principle, since you will have to look for solutions in both directions, choosing some odd number as the start or finish, but achievable solutions will still be limited to only a part of all nodes of such a network — a sub-tree when searching upward and an infinite descent trajectory when searching downwards.

To summarize: with the Collatz conjecture, if not everything, then a lot is already fairly clear. I guess it is time to rethink both the unsolved problem itself and the long, tangled history of attempts to solve it.

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